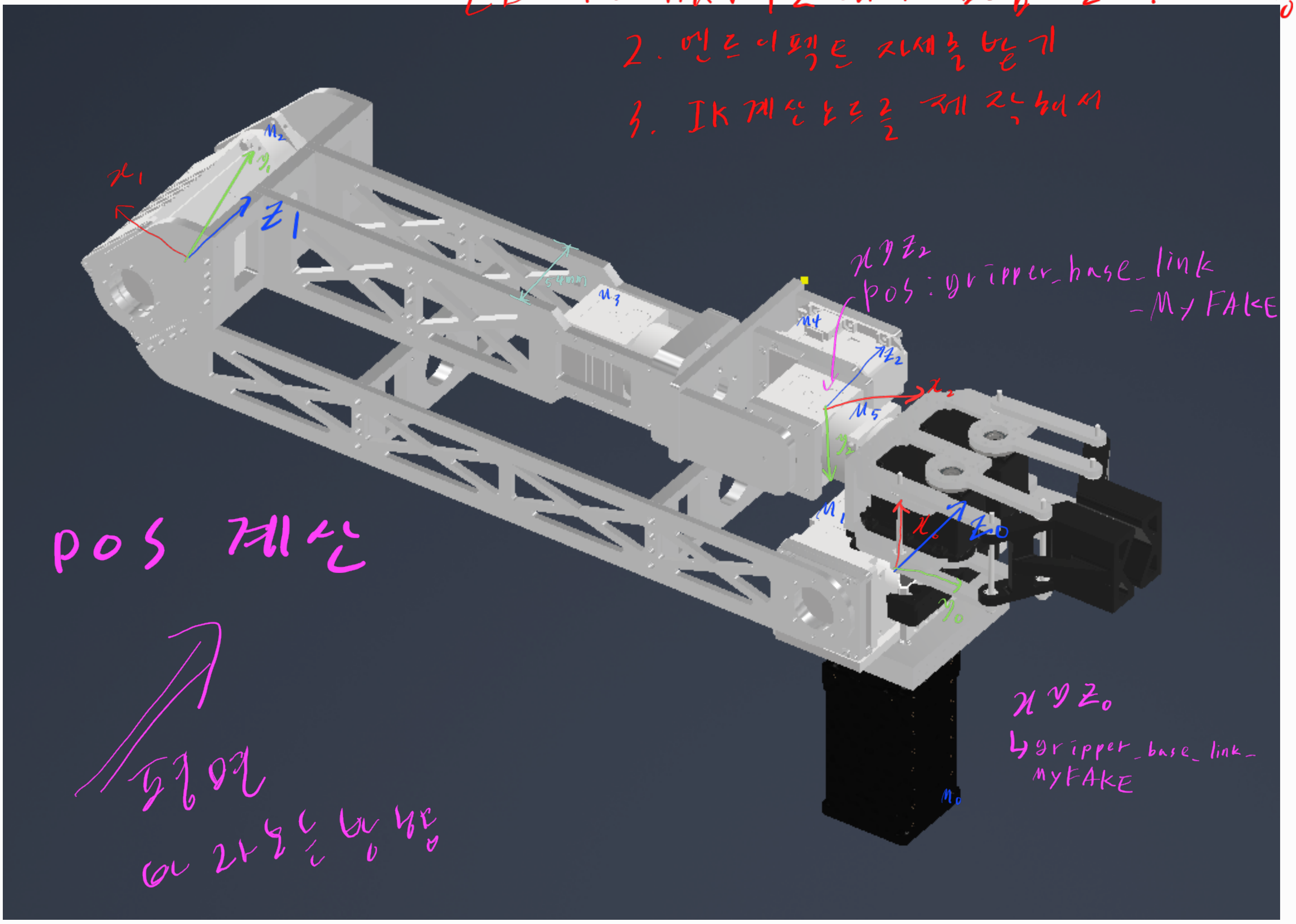
과제 : 1.Rviz 에서 마음대로 위치이동
2. 엔드이펙트 자세를 받기
3. IK 계산노드를 제작해서



xyz_2
pos : gripper_base_link_MyFAKE

xyz_0
↳ gripper_base_link_myFAKE

pos 계산

평면으로 자르는 방법

1. 좌표를 평면으로 잘라서 2링크로 해석. (M_1, M_2 계산)

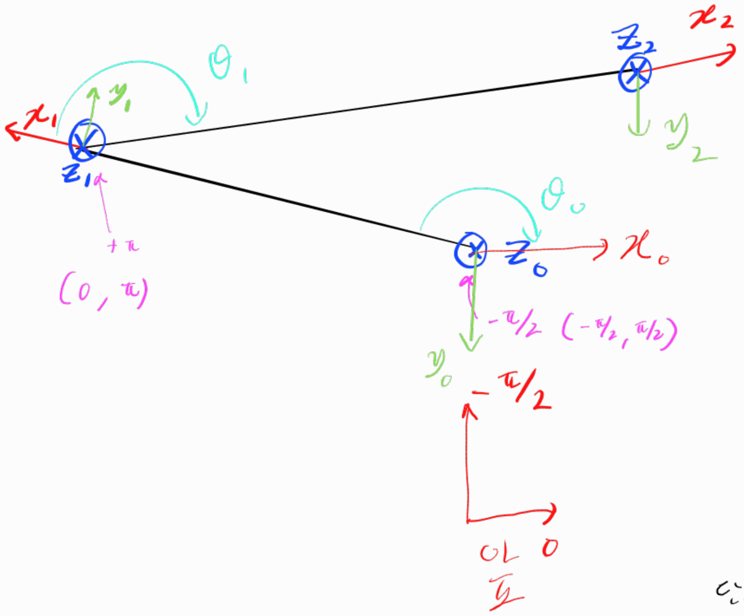
2. 좌표평면 tan 으로 M_0 구하기

3. 회전행렬과 Pitch-Roll-Yaw, 역행렬(전치행렬)을 이용해
M_{345} 구하기

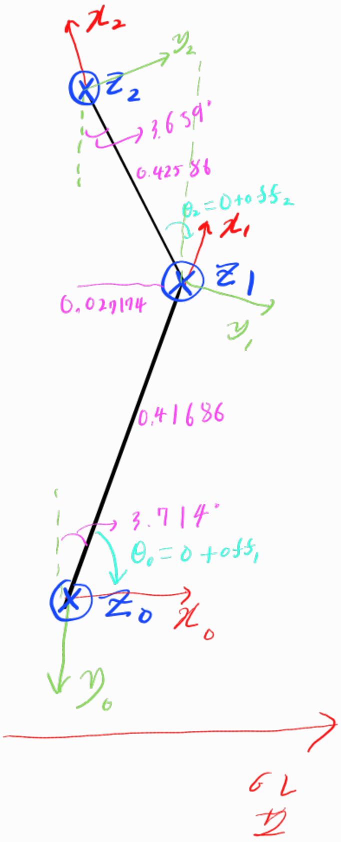
4. 필요에따라 Rviz 의 .xacro를 조작하여 링크를 추가한다.

★ 인벤터랑 RVIZ 간 수치가 일부 달라서
RVIZ로 수정 계산 함

정기구학 (RVIZ)



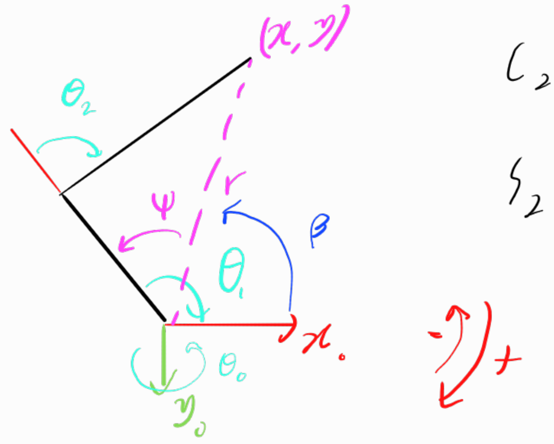
RVIZ 원점



단위: m, deg

	d_i	θ_i	a_i	α_i
1	0	$0 - 86.286°$	0.41686	0
2	0	$0 - 7.373°$	0.42586	0

도리선 계산 : 코드(2)

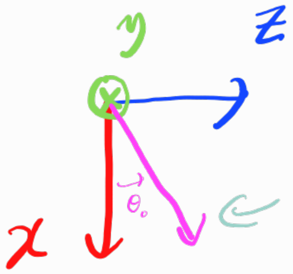


$$c_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$s_2 = \pm\sqrt{1 - c_2^2}$$

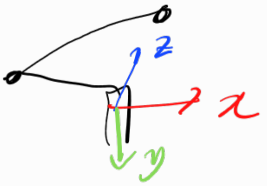
$$\theta_2 = \mathrm{atan2}(s_2, c_2)$$

$$\theta_1 = \mathrm{atan2}(y, x) - \mathrm{atan2}(a_2s_2, a_1 + a_2c_2)$$

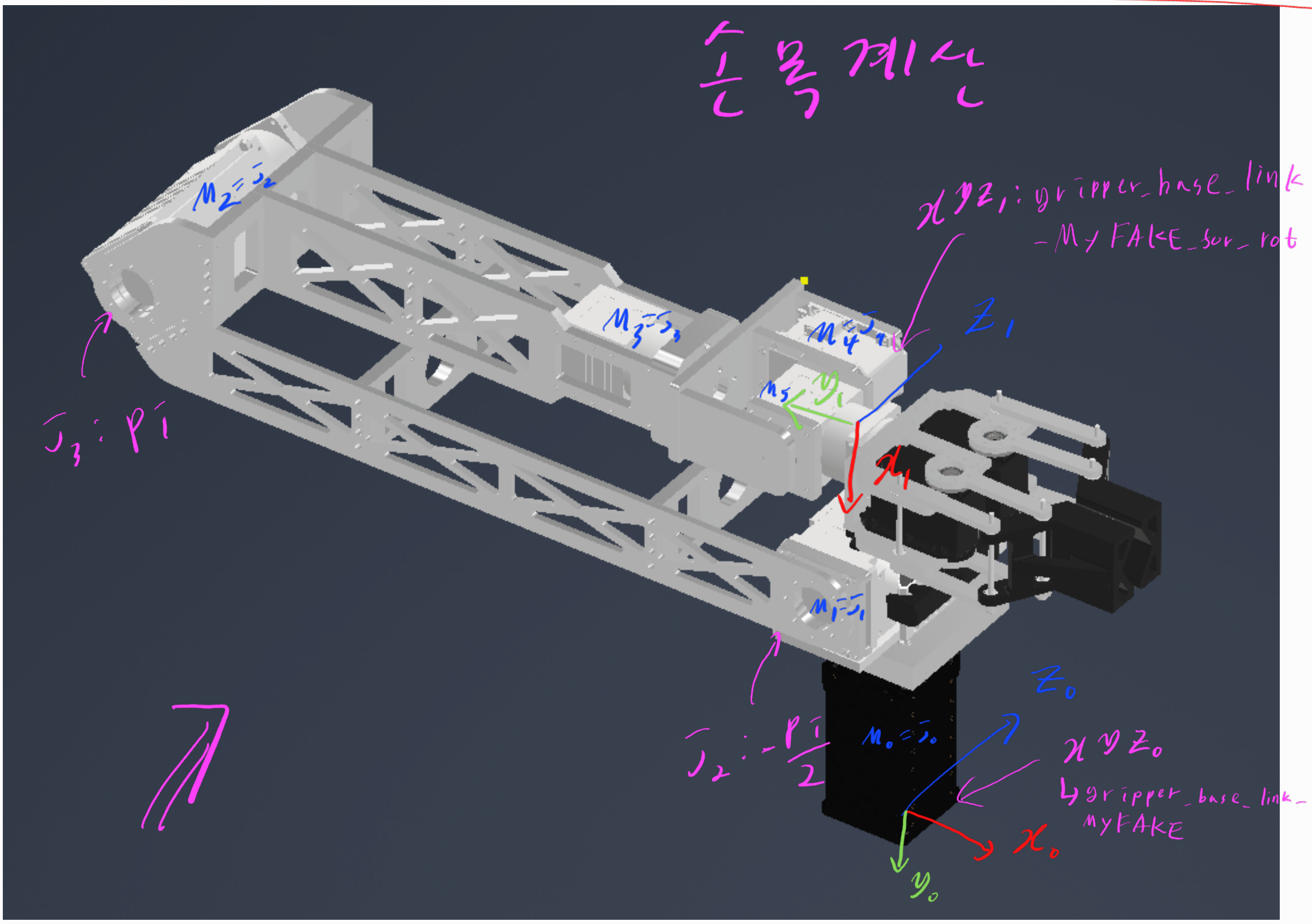


$\theta_0 = \mathrm{atan2}(z, x)$

← 도형면 기준: 코드①

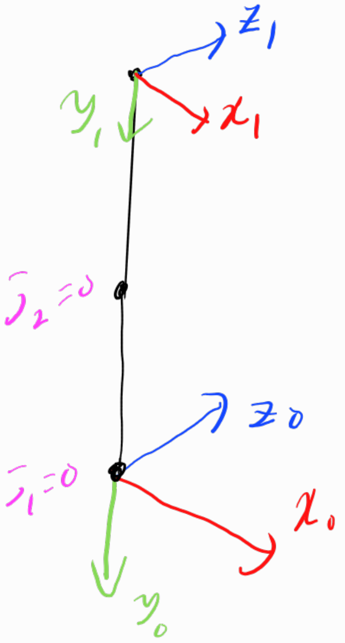


손목계산



xyz_1: gripper_base_link_MyFAKE_sor_rot

xyz_0 ↳ gripper_base_link_MyFAKE



$$R_{목표} = R_{팔} \cdot R_{손목}$$

$$R_{손목} = R_{팔}^{-1} \cdot R_{목표}$$

$\searrow R_{팔}^{T}$

$$R_{팔} = R_{J_0(M_0)} \cdot R_{J_1(M_1)} \cdot R_{J_2(M_2)}$$

$$= [-\mathrm{Pitch}(J_0)] \cdot [+\mathrm{Yaw}(J_1)] \cdot [+\mathrm{Yaw}(J_2)]$$

$$R_{손목} = [-P(\bar{J}_3)] \cdot [Y(\bar{J}_4)] \cdot [-P(\bar{J}_5)]$$

$$-\bar{J}_3 = a, \quad \bar{J}_4 = b, \quad -\bar{J}_5 = c$$

$$= \begin{bmatrix} ca & 0 & sa \\ 0 & 1 & 0 \\ -sa & 0 & ca \end{bmatrix} \cdot \begin{bmatrix} cb & -sb & 0 \\ sb & cb & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} cc & 0 & sc \\ 0 & 1 & 0 \\ -sc & 0 & cc \end{bmatrix}$$

$$= \begin{bmatrix} ca \cdot cb \cdot cc - sa \cdot sc & \boxed{-ca \cdot sb} & ca\,cb\,sc \cdot sa\,cc \\ \boxed{sb \cdot cc} & \boxed{cb} & \boxed{sb \cdot sc} \\ -sa \cdot cb \cdot cc - ca \cdot sc & \boxed{sa \cdot sb} & -sa \cdot cb \cdot sc + ca \cdot cc \end{bmatrix}$$

$$\begin{array}{ccc} b, c & b & a, b \end{array}$$

$$\bar{J}_4 = b = \operatorname{atan2}\left(\overset{이거\ 사용}{\pm}\sqrt{R_{12}^2 + R_{32}^2},\ R_{22}\right)$$

$$\bar{J}_3 = -a = -\operatorname{atan2}(R_{32}, -R_{12})$$

$$\bar{J}_5 = -c = -\operatorname{atan2}(R_{23}, R_{21})$$